

Project Documentation

Software Installation Request Automation using ServiceNow Service Catalog

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Team ID	Individual / Solo Project
Platform	ServiceNow Personal Developer Instance (PDI)

1. Introduction

This document describes the step-by-step implementation of the Software Installation Request Automation solution on a ServiceNow Personal Developer Instance (PDI). It covers configuration of the Business Rule, Workflow, Service Catalog item, catalog variables, end-to-end testing, and migration of the solution via Update Sets — with screenshots captured directly from the PDI at each stage.

2. Business Rule Configuration

A Business Rule was configured to automatically raise an incident when a software request cannot be fulfilled due to a license shortage (i.e., when the request state changes to On Hold).

Steps

1. Open ServiceNow.
2. Search for Business Rules.
3. Click on New.
4. Fill in the details as shown below.

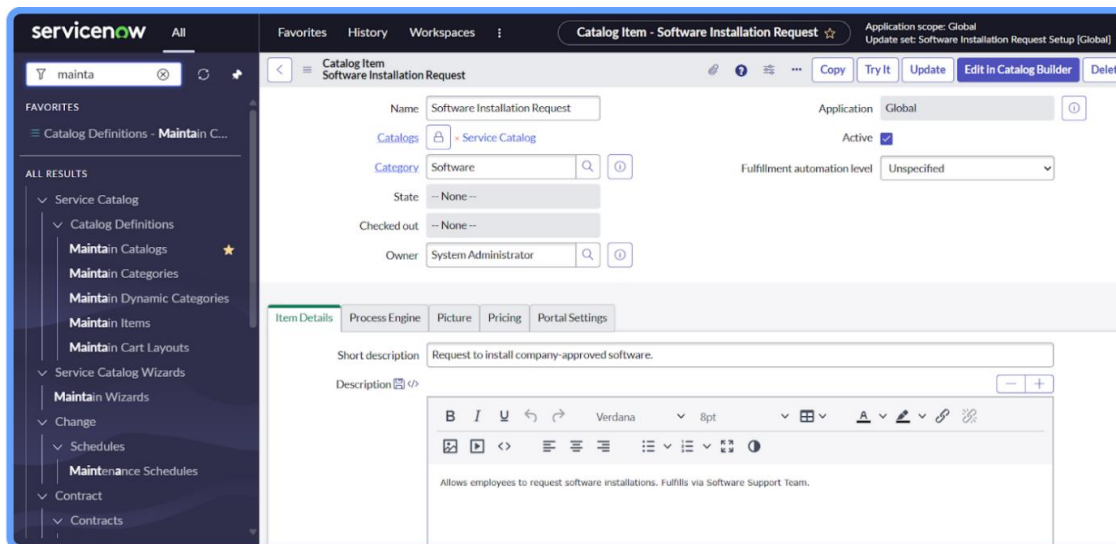
The screenshot shows the ServiceNow interface for creating a new Catalog Item. The left sidebar displays the 'Catalog Definitions - Maintain C...' menu with various options like 'Maintain Catalogs', 'Maintain Categories', etc. The main form is titled 'Catalog Item - Software Installation Request'. It includes fields for 'Name' (Software Installation Request), 'Application' (Global), 'Category' (Software), 'State' (None), 'Checked out' (None), and 'Owner' (System Administrator). There are also buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. The 'Short description' field contains 'Request to install company-approved software.' and the 'Description' field contains 'Allows employees to request software installations. Fulfills via Software Support Team.' The form is divided into tabs: 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'.

Figure 1: Creating a new Business Rule

5. Click on Advanced.

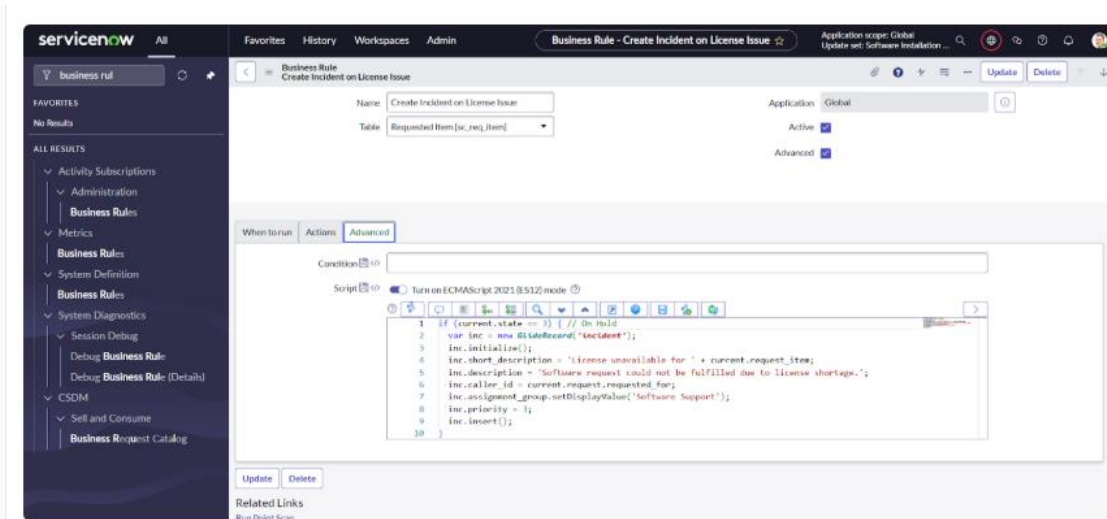


Figure 2: Advanced tab of the Business Rule

Condition: [State] [changes to] [On Hold] (or a custom field such as Issue = True)

Script logic used in the Business Rule:

```
if (current.state == 3) { // On Hold
    var inc = new GlideRecord('incident');
    inc.initialize();
    inc.short_description = 'License unavailable for ' + current.request_item;
    inc.description = 'Software request could not be fulfilled due to license shortage.';
    inc.caller_id = current.request.requested_for;
    inc.assignment_group.setDisplayValue('Software Support');
    inc.priority = 3;
    inc.insert();
}
```

6. Click on Save/Submit.

3. Workflow Creation

3.1 Creating the Workflow

Steps:

- Open ServiceNow.
- Click on All >> search for Workflow.

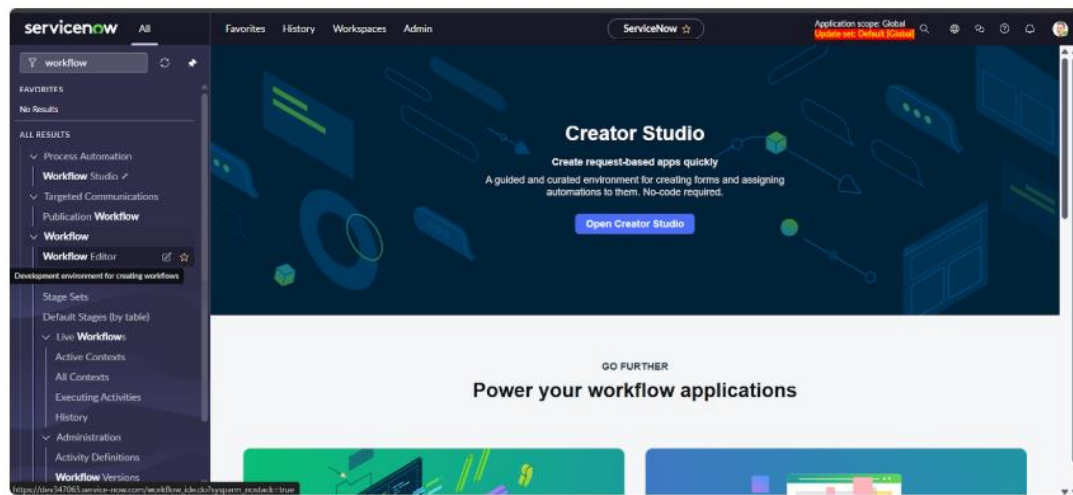


Figure 3: Navigating to the Workflow module

- Click on Workflow Editor.
- Click on New Workflow.

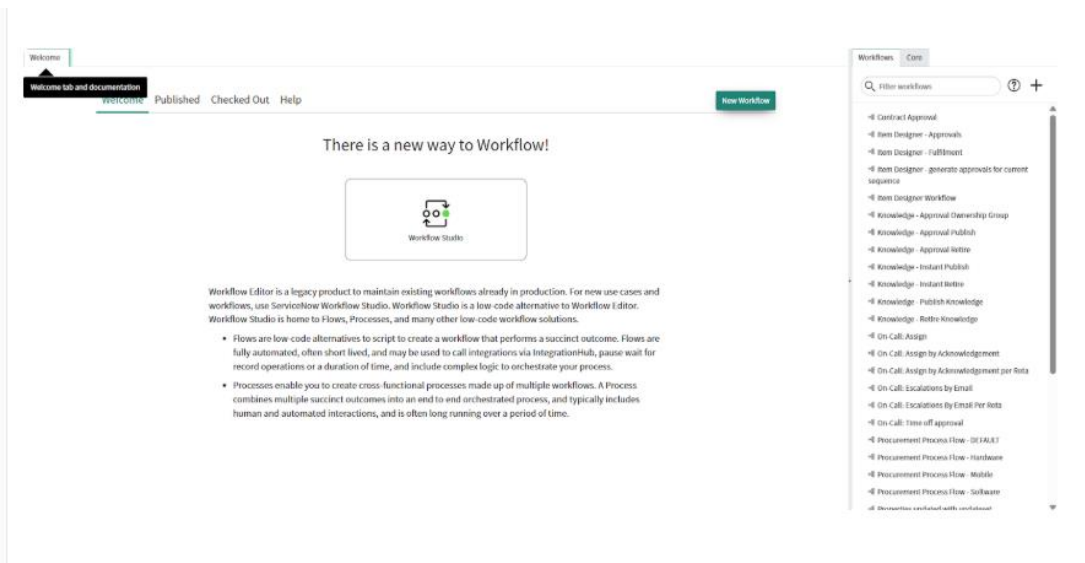


Figure 4: Creating a new Workflow

- Fill in the details as shown below.
- Click on Submit.

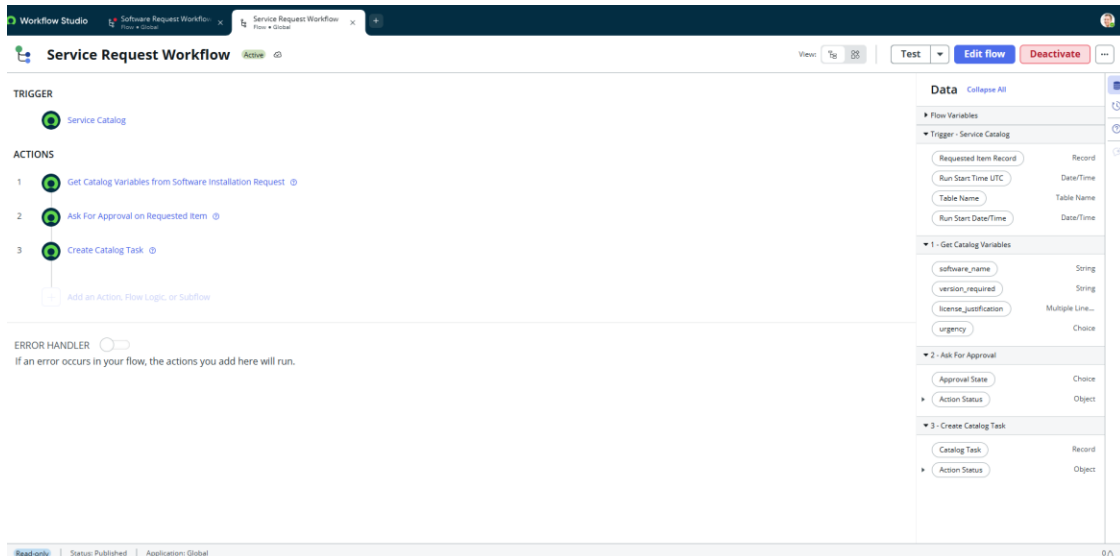


Figure 5: Workflow submission details

3.2 Designing the Workflow

1. Design the Workflow (default).
2. Add Approval Step (Optional).

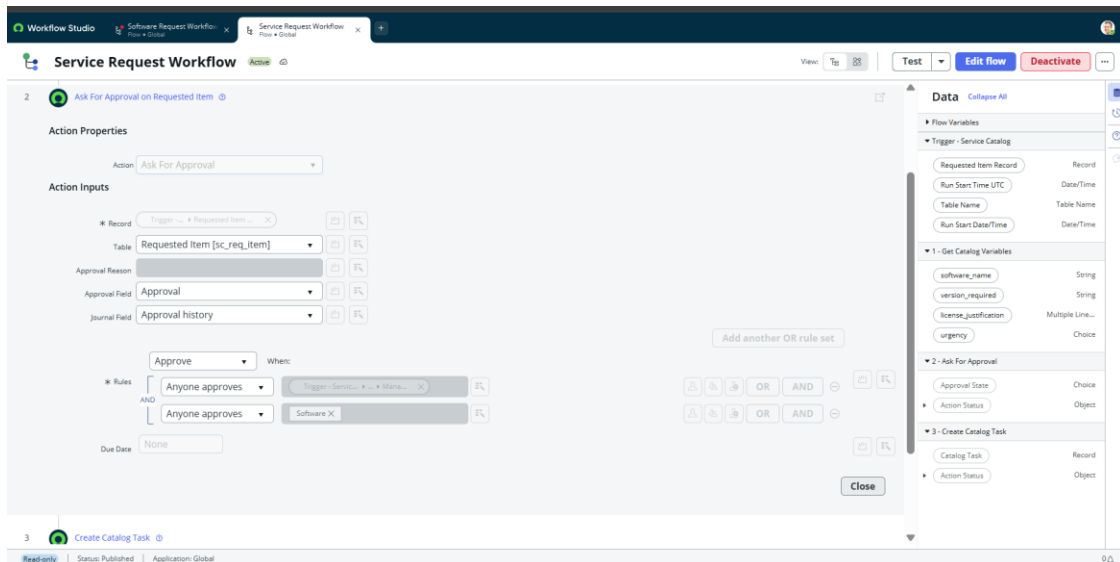


Figure 6: Workflow canvas with Approval activity

Click on Core on the right side and drag the Approval Group under the Approvals section.

3. Fill in the details as given below.
4. Do the same process to create the task from the 'Tasks' section.

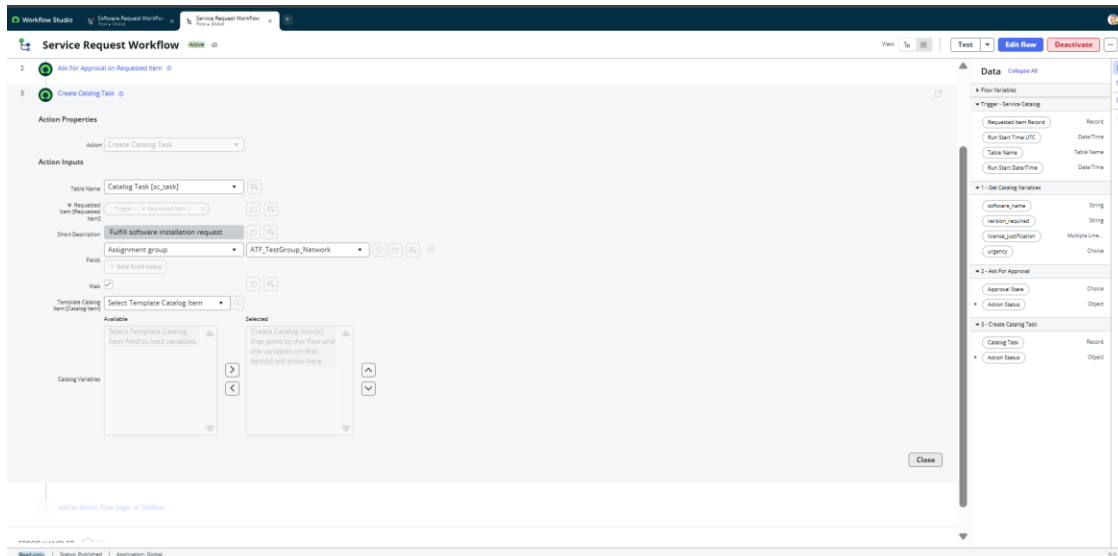


Figure 7: Adding a Task activity to the workflow

5. Add an End activity — drag it to the canvas to mark workflow completion.
6. Connect the activities in sequence.
7. Click on Publish (top left of the Workflow Editor).

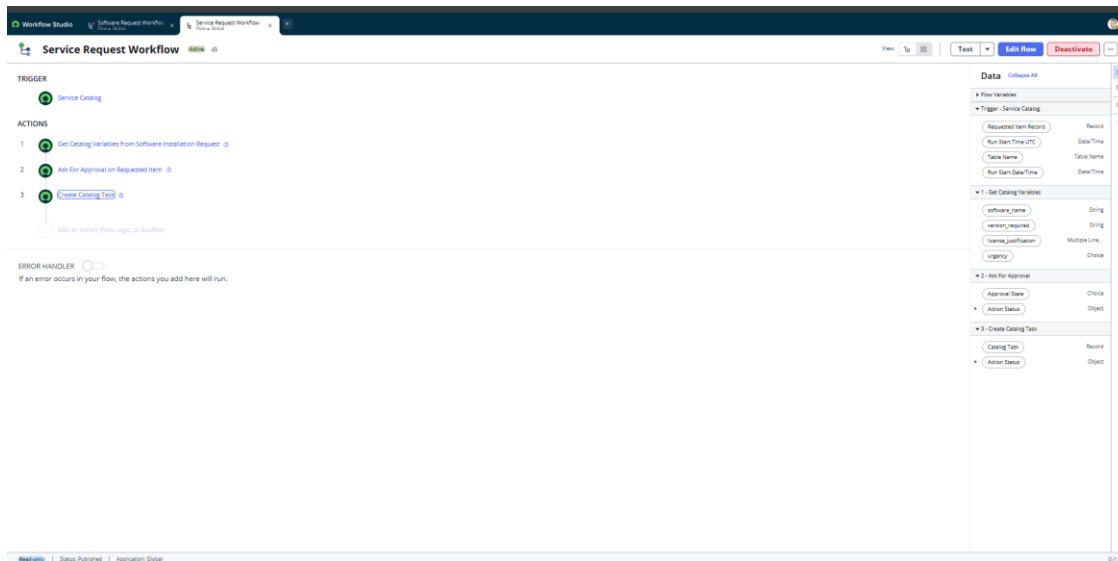


Figure 8: Publishing the completed workflow

4. Service Catalog Item Creation

4.1 Creating the Catalog Item

- Open ServiceNow.
- Click on All >> search for Maintain Items.
- Click on New and fill in the details as shown below.

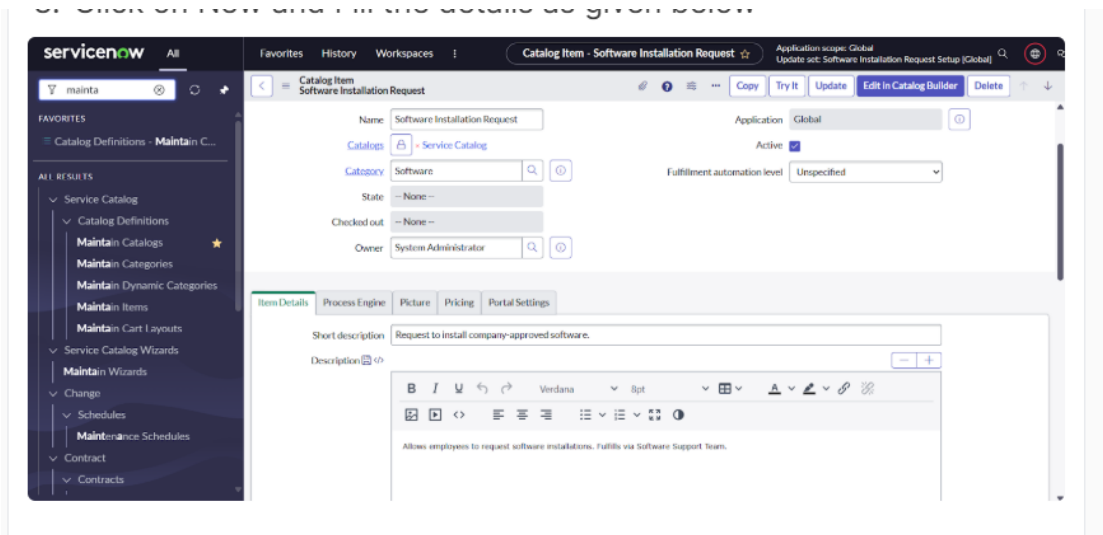


Figure 9: Creating a new Service Catalog item

Give a suggestion name in the Meta field so that the item can be searched easily.

4.2 Variables Configuration

After saving the catalog item, scroll down and click on Variables. Click New and create four variables as follows:

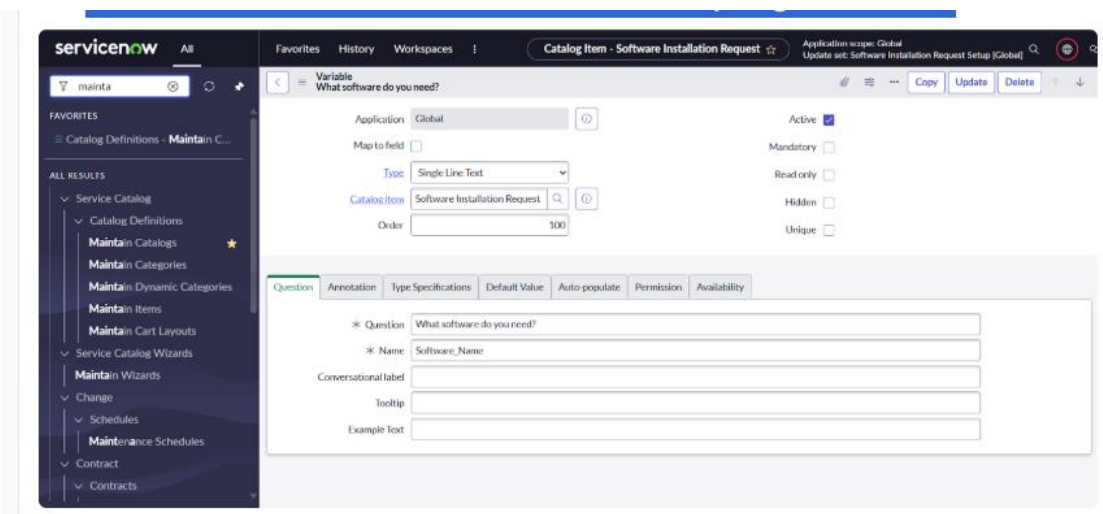


Figure 10: Adding catalog variables

- Variable 1: Software Name — Type: Single Line Text — Question: What software do you need?
- Variable 2: Version Required — Type: Single Line Text — Question: Specify version (if required).
- Variable 3: License Justification — Type: Multi Line Text — Question: Why do you need this software?
- Variable 4: Urgency — Type: Choice (Options: Normal, High, Critical) — Question: Select urgency level.

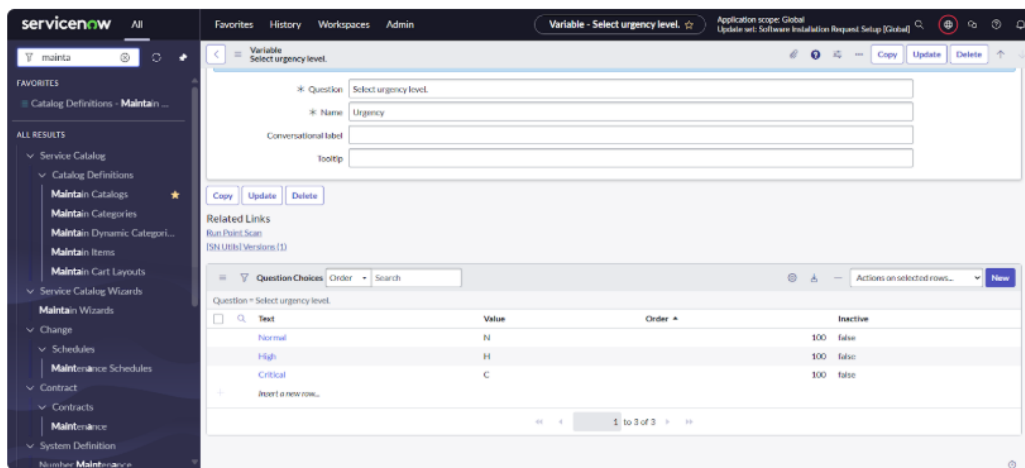


Figure 11: Configured catalog variables

Click on Save.

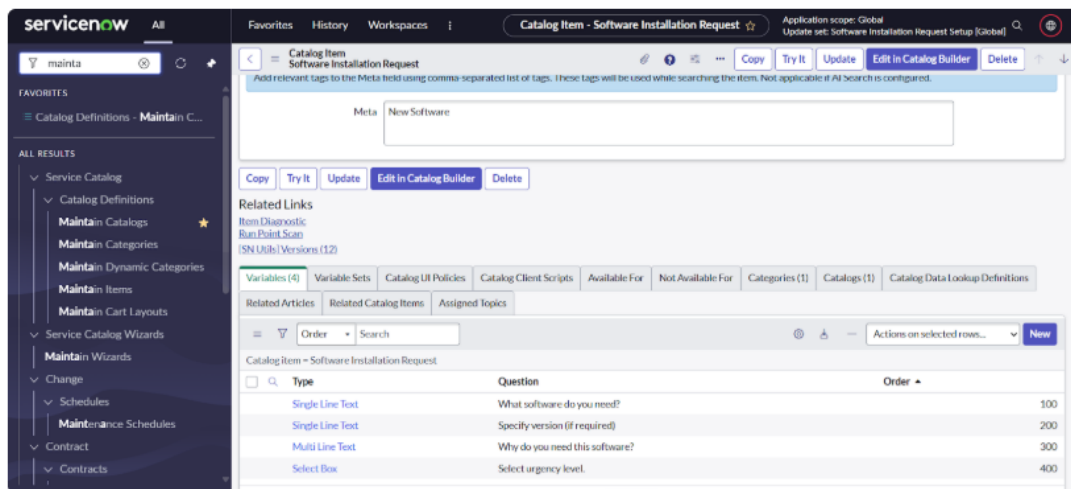


Figure 12: Saved catalog item with variables

5. Navigation Flow

Click on Try It to preview the variables that were created, then fill in the details as shown below.

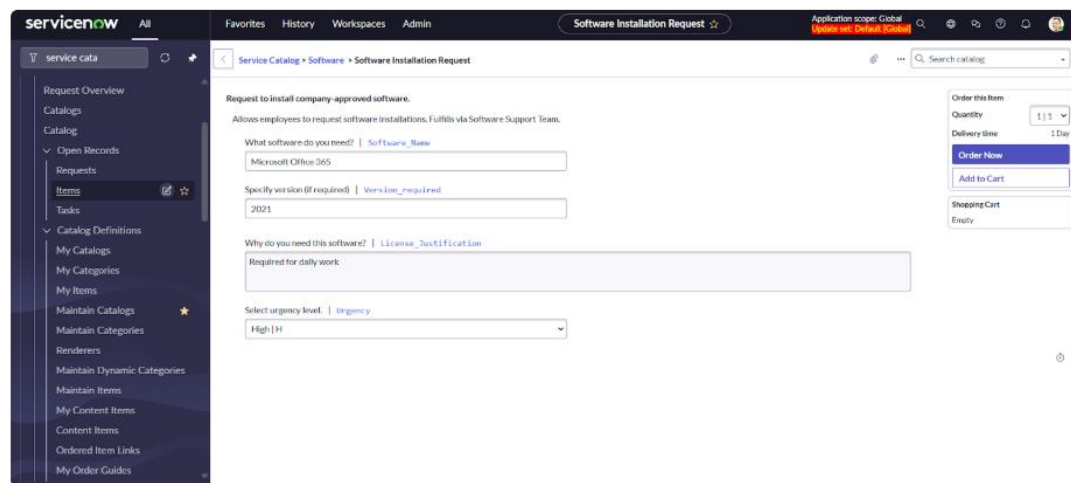


Figure 13: Previewing the catalog item ('Try It')

Click on Order Now.

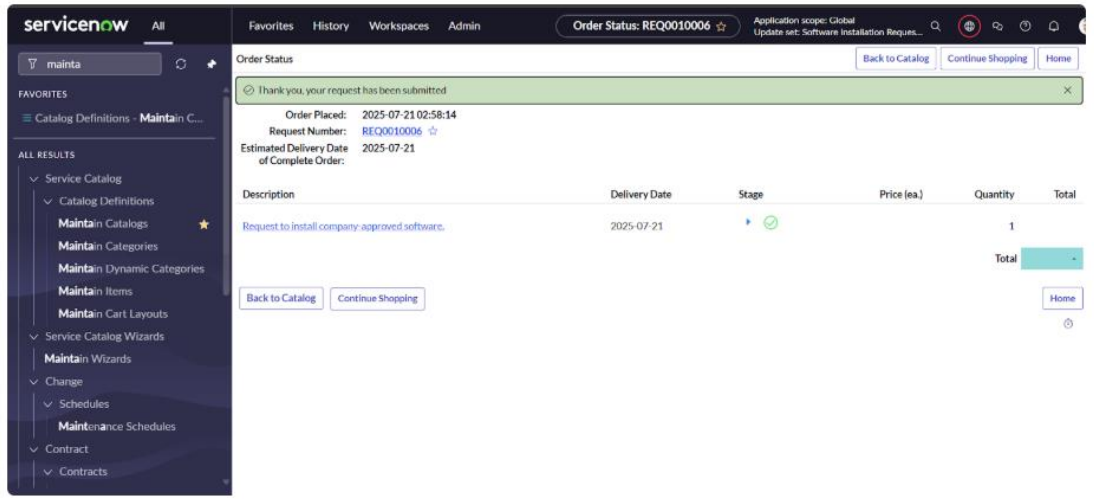


Figure 14: Placing an order for the catalog item

6. End-to-End Testing

Test 1: Network Requests Procedure

1. Log in to the ServiceNow PDI.
2. Copy the instance domain, e.g. <https://dev314490.service-now.com>.
3. Paste the URL in a new tab and add the prefix 'sp', e.g. <https://dev314490.service-now.com/sp>.
4. Search for Network Requests.
5. Fill in the required details and click Submit.

A new request is generated with a request number, and users receive a corresponding confirmation email.

Test 2: Testing Catalog Variables in Service Portal

Open the Service Portal and search for 'New Software' (the same name given in the Meta field earlier). Fill in the details as shown below, then click Order Now.

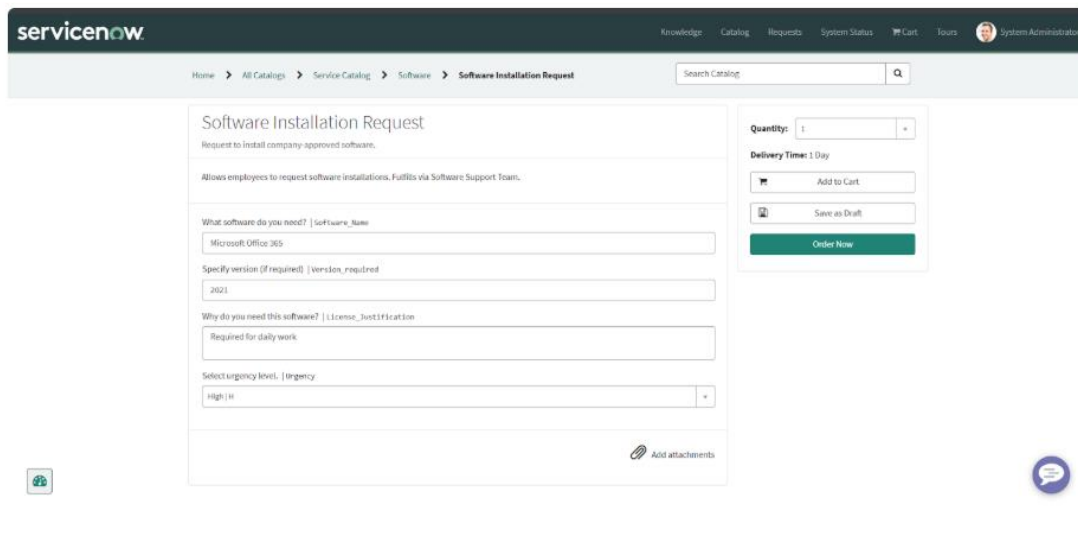


Figure 15: Submitting the software request via Service Portal

Click on Check Out.

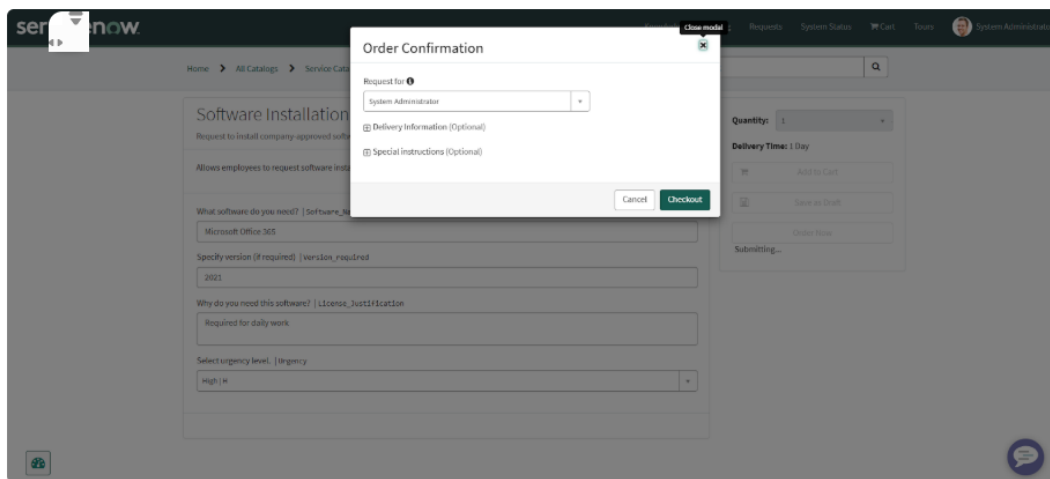


Figure 16: Checkout confirmation screen

Usability — Order Confirmation

The order number (record number) is displayed after checkout.

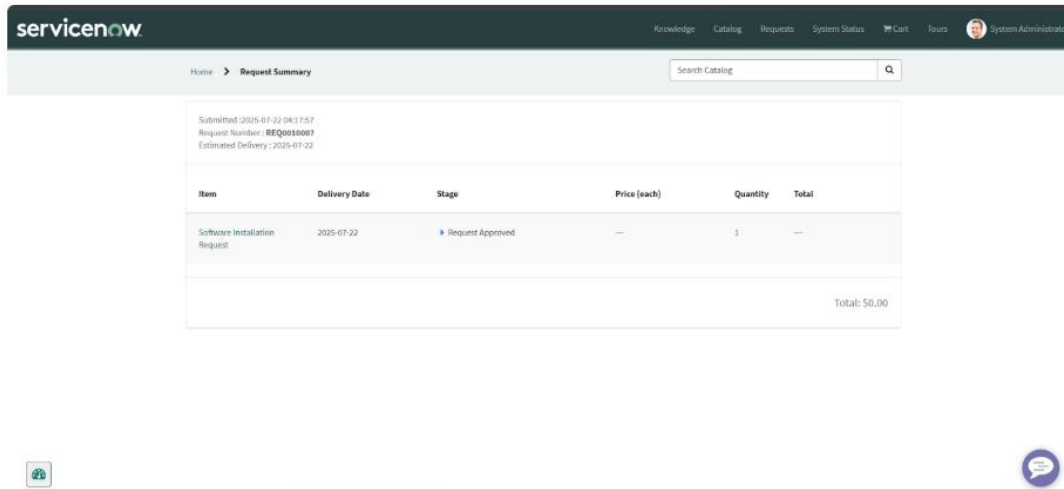


Figure 17: Order confirmation with record number

Tooltips & Help Text

Tooltip/help text icons were added next to fields to guide users who already have a license.

Software Installation Request
Request to install company approved software.

Allows employees to request software installations. Fulfills via Software Support Team.

What software do you need? | Software_name
Microsoft Office 365

Specify version (if required) | Version_required
2021

Why do you need this software? | License_justification
Required for daily work

Select urgency level. | urgency
High (H)

Add attachments

Quantity: 1

Delivery Time: 1 Day

Add to Cart

Save as Draft

Order Now

Figure 18: Tooltip and help text on catalog fields

Activity: Adding the Workflow to the Catalog Item

- Open the newly created Catalog Item.
- Click on Process Engine.
- Add the Software Request Workflow in the workflow field.
- Click on Save.

ServiceNow All Favorites History Workspaces Catalog Item - Software Installation Request

Application scope: Global
Update set: Software Installation Request Setup [Global]

Copy Try It Update Edit in Catalog Builder Delete

Catalogs: Service Catalog

Category: Software

State: --None--

Checked out: --None--

Owner: System Administrator

Active: ☒

Fulfillment automation level: Unspecified

Item Details Process Engine Picture Pricing Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

Flow:

Workflow: Software Request Workflow

Execution Plan:

Copy Try It Update Edit in Catalog Builder Delete

Related Links
Item Diagnostic
Run Policy Scan

Figure 19: Attaching the workflow to the catalog item

Test 3: Testing Catalog Requests, Items, and Tasks

sc_request Table (Service Catalog Request Table): Open ServiceNow, search for Service Catalog > Requests, and locate the Request Number (e.g., REQ0010006).

The screenshot shows a ServiceNow Request record for REQ0010006. The left sidebar contains navigation links for Service Catalog, Catalog Builder, Request Overview, Catalogs, Open Records, Requests, Items, Tasks, Catalog Definitions, My Catalogs, My Categories, My Items, Maintain Catalogs, Routers, Maintain Dynamic Catalogs, and Maintain Items. The main form displays the following fields:

- Number: REQ0010006
- Requested for: System Administrator
- Location: [empty]
- Due date: 2025-07-21 16:58:13
- Price: \$0.00
- Description: [empty]
- Short description: [empty]
- Special instructions: [empty]
- Opened: 2025-07-21 02:58:54
- Opened by: System Administrator
- Approval: Approved
- Request state: Approved

Buttons at the bottom include Update, Cancel Request, Copy, and Delete. Below the form is a 'Related Links' section with 'Show Workflow' and 'Workflow Content'. A table at the bottom shows 'Requested Items (1)' with one entry:

Number	Quantity	Catalog	Item	Due date	Price	Assigned to	Stage
RITM0010011	1 (empty)		Software Installation Request	2025-07-21 16:58:13	\$0.00 (empty)		Open

Figure 20: Service Catalog Request record

Test 4: sc_req_item Table (Requested Item Table)

Under Service Catalog Requests, click on Items, then open the request item number. The catalog item variables with their backend names are auto-filled.

The screenshot shows a ServiceNow Requested Item (RITM0010011) record. The left sidebar is the same as in Figure 20. The main form displays the following fields:

- Number | number: RITM0010011
- Item | cat_item: Software Installation Request
- Request | request: REQ0010006
- Requested for | requested_for: System Administrator
- Due date | due_date: 2025-07-21 16:58:13
- Configuration item | configuration_item: [empty]
- Watch list | watch_list: [empty]
- Opened | opened_at: 2025-07-21 02:58:54
- Opened by | opened_by: System Administrator
- State | stage: Request Approved | request_approved
- State | state: Open | 1
- Quantity | quantity: 1
- Estimated delivery | estimated_delivery: [empty]
- Backordered | backordered: [empty]
- Order Guide | order_guide: [empty]

Below the form is a 'Variables' section with the following fields:

- What software do you need? | Software_Name: Microsoft Office 365
- Specify version (if required) | Version_required: 2021
- Why do you need this software? | License_justification: Required for daily work
- Select urgency level | urgency: High

Buttons at the bottom include Additional comments (Customer) and Additional comments (Customer visible).

Figure 21: Requested Item (RITM) record with variables

Activity: sc_task Table (Catalog Task Table)

The same process is repeated for the task table. Verify that:

- The task is assigned to the Software Support Team.
- The short description is populated correctly (e.g., 'Install Microsoft Office').
- The task state reflects progress (Open, Work in Progress, Completed).

7. Finalizing and Moving the Update Set

Open the Update Set that was created during development and change its state from In Progress to Complete, then click Update.

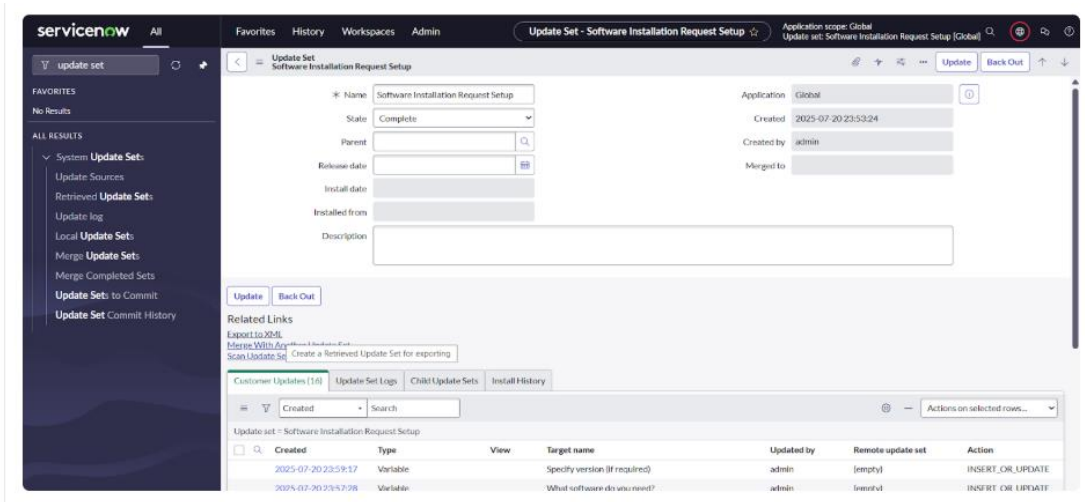


Figure 22: Completing the Update Set

Click on Export to XML and download the file.

8. Testing the Update Set on Another Instance

Open a second ServiceNow instance and search for Retrieve Update Sets, then click Export Update Set from XML.

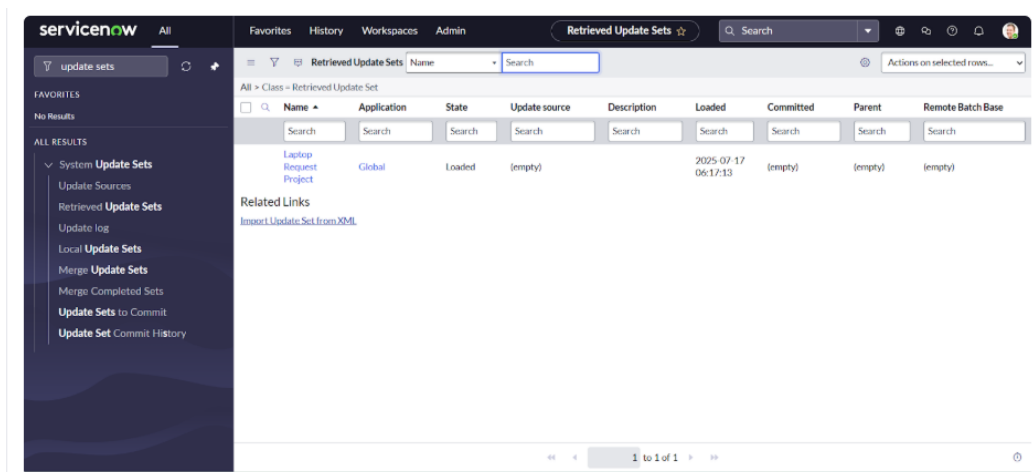


Figure 23: Retrieve Update Sets screen

Select the file downloaded from the source instance and click Upload.

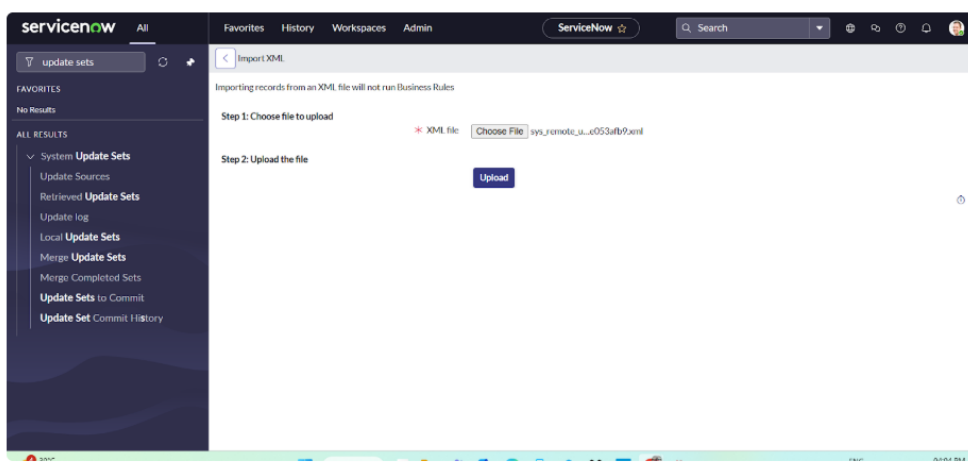


Figure 24: Uploading the Update Set XML file

After uploading, the new record appears in the list.

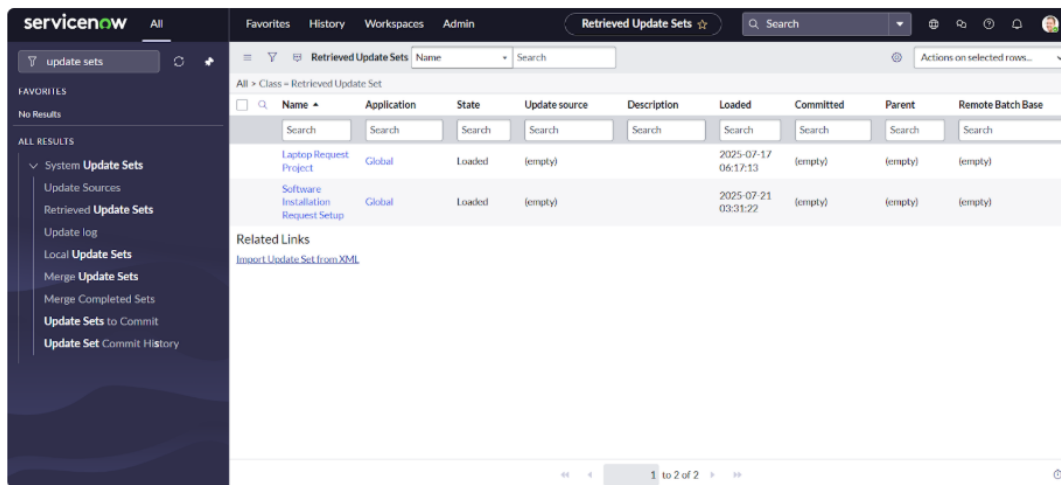


Figure 25: Uploaded Update Set record

Open the record to review its contents.

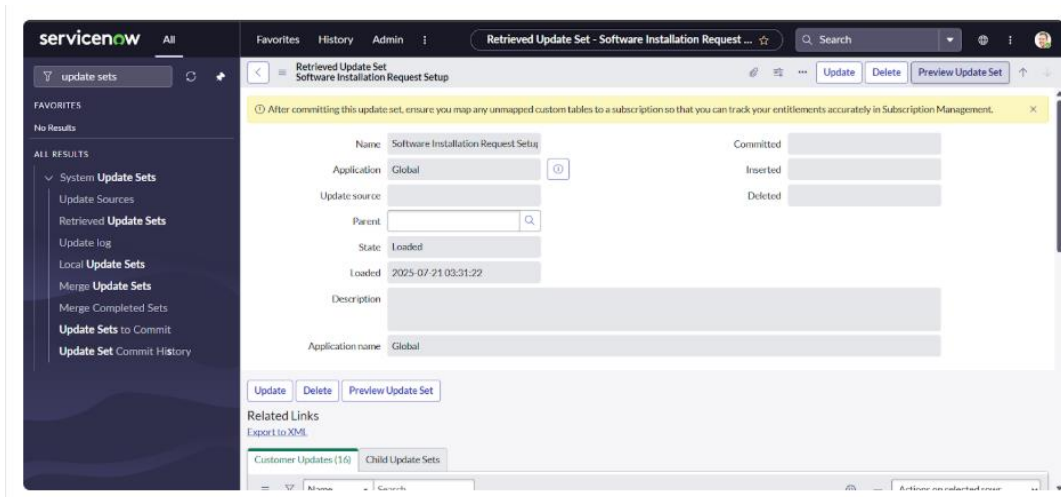


Figure 26: Reviewing Update Set contents

Click on Preview Update Set.

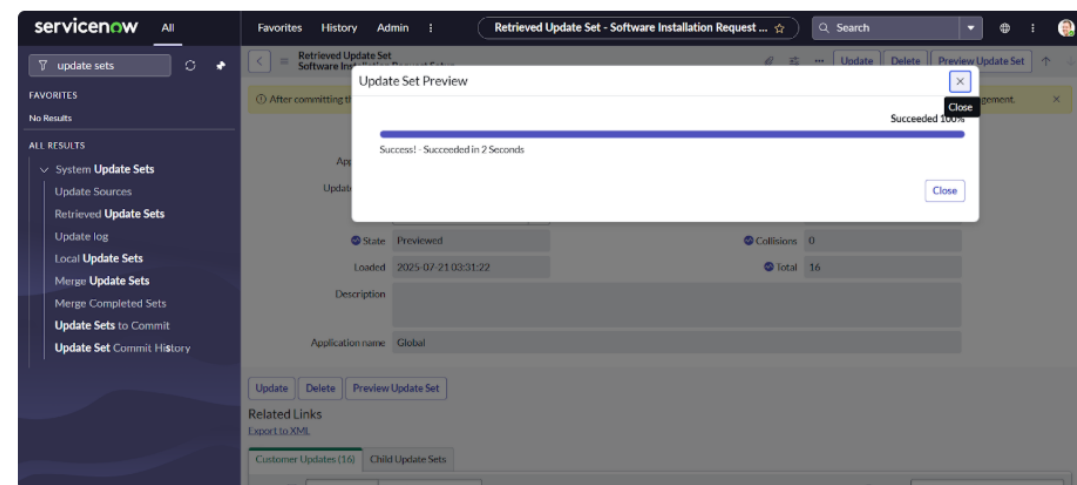


Figure 27: Previewing the Update Set before commit

Click on Commit Update Set to transfer all the data.

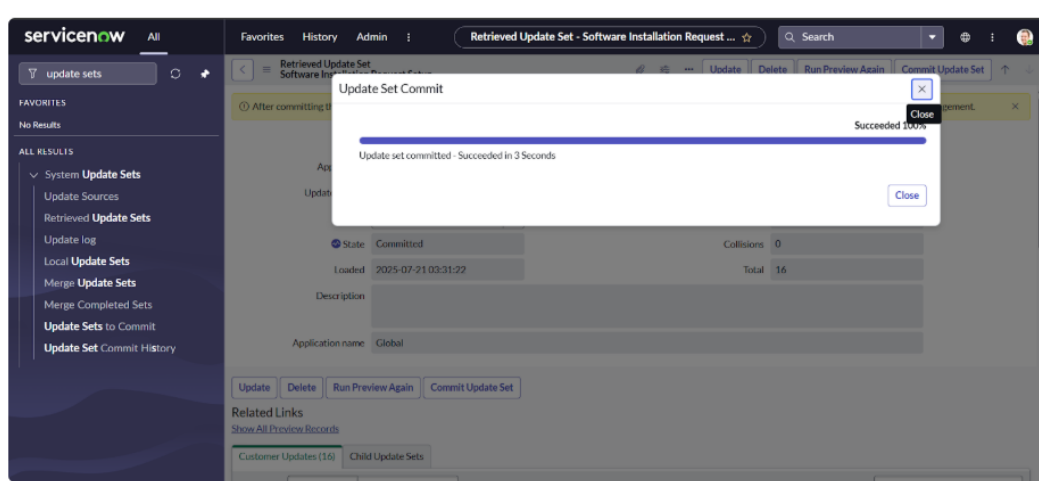


Figure 28: Committing the Update Set

All data is now transferred to the target instance. Validate that no elements are missing, and re-run all the activities and tests on this instance to confirm successful migration.

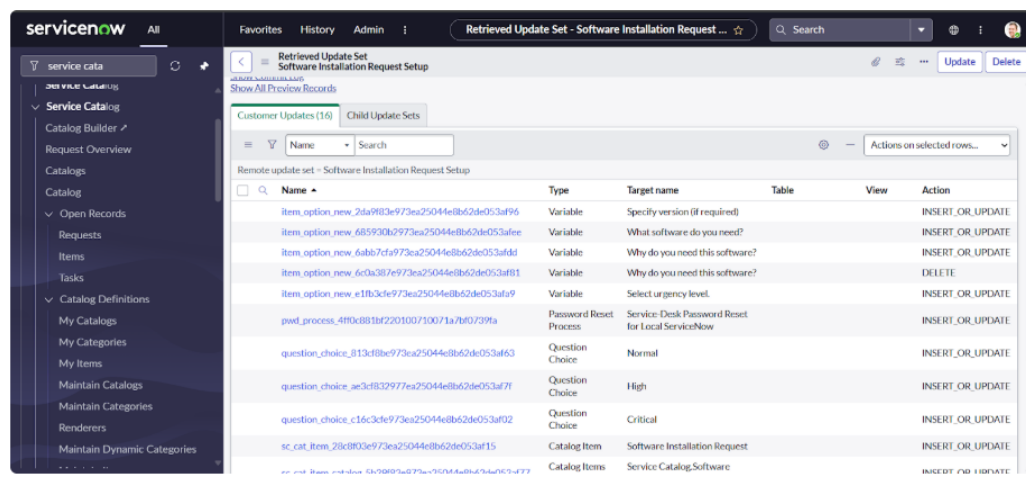


Figure 29: Confirmed data transfer on the target instance

9. Conclusion

This documentation walks through the complete build of the Software Installation Request Automation solution — from Business Rule and Workflow configuration, through Service Catalog item and variable setup, to full end-to-end testing across the request, item, and task tables — and confirms successful migration of the solution between ServiceNow instances using Update Sets.